

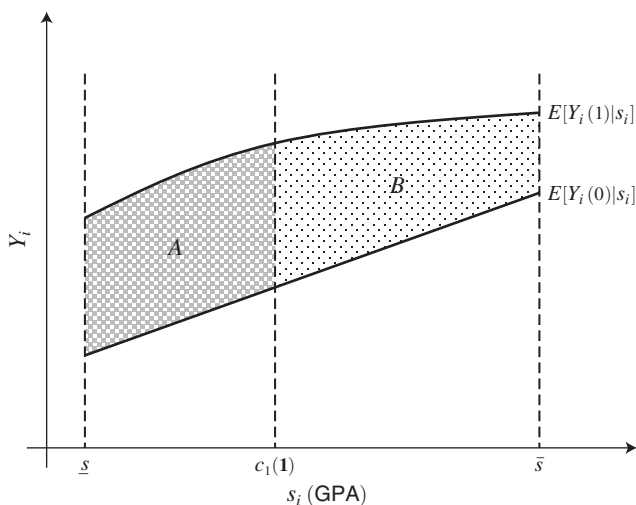


FIGURE 2. TOTAL POTENTIAL GAINS AND LOSSES FROM THE REFORM: A THEORETICAL ILLUSTRATION

Notes: $E[Y_i(1)|s_i]$ indicates the mean outcome for students with GPA s_i who are externally assigned to the competitive school, and $E[Y_i(0)|s_i]$ indicates the mean outcome for the same students if they are externally assigned to the noncompetitive school. Assume that s_i is uniformly distributed over the interval $[s, \bar{s}]$. When the admission scheme becomes merit based, only students with GPA $s_i \geq c_1(\mathbf{1})$ are able to switch to the competitive school, with the total potential gain from attending the competitive school equal to area B on the diagram. Similarly, area A indicates the total potential loss from being excluded from attending the competitive school after the reform. See Section II for details.

the noncompetitive school. We make no restrictions on the mean conditional outcomes; the difference between $E[Y_i(1)|s_i]$ and $E[Y_i(0)|s_i]$ can vary flexibly by student's GPA. When the admission scheme becomes merit based, only students with GPA $s_i \geq c_1(\mathbf{1})$ can switch to the competitive school, with the total potential gain from attending the competitive school equal to area B on the diagram.¹³ The actual (or realized) gain will depend on the mass of students who are above the admission cutoff ($s_i \geq c_1(\mathbf{1})$) and switch to the competitive school as a result of the reform, which is determined by their priorities (and hence locations) before the reform, as well as their preferences. Provided that they weakly prefer the competitive school to the noncompetitive school, the model implies that the fraction of students attending the competitive school should increase after the reform.¹⁴ This is a key insight that we build on in our identification strategy (see Section IIIB). On the contrary, low-achieving students with $s_i < c_1(\mathbf{1})$ may be crowded out from the competitive school, leading to a total potential loss that is equal to area A on the same diagram. The aggregate impact therefore depends on the relative size of A and B , as well as

¹³In this example, assume that s_i is uniformly distributed over $[s, \bar{s}]$.

¹⁴For these high-ability students, this preference ordering is required to rule out the case that they might switch to the noncompetitive school under the merit-based system (these would be high-ability students living in the catchment area for the competitive school but opt for the noncompetitive school under the merit-based system). To justify this assumption, in Sections I and VD, we provide comprehensive evidence documenting that competitive schools outperformed noncompetitive schools even before the reform, in terms of characteristics such as teacher quality, average student outcomes and value added. Note that we do not need to impose any preference ordering for students below the admission cutoff.